Partha Sarathi Purkayastha

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Research Interests

AI in Education, AI in Healthcare, Human-AI Interaction, Safety & Security, Agentic AI, LLM Evaluation

Education

2025 – 2027 **M.Sc. Computer Science**, ETH Zürich, Zürich, Switzerland
Relevant Coursework: Probabilistic Artificial Intelligence, Large Language Models, Mixed Reality, AI in Education, Big Data, Information Security Lab, Computational Intelligence Lab, Computational Statistics

2020 – 2024 **B.E. Computer Science**, Birla Institute of Technology and Science, Pilani, India
CGPA: 9.83/10, **Director's Gold Medal Awardee**
Institute Merit Scholarship: Awarded a full tuition waiver for consistently ranking in the **top 1%**.
Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming, Database Systems, Theory of Computation, Operating Systems, Computer Networks, Linear Algebra, Probability and Statistics, Discrete Mathematics, Artificial Intelligence, Computer Graphics

Accepted & Published

AIED 2026 **Partha Sarathi Purkayastha**. Reliability as a Teammate: Budgeted Verifier-in-the-Loop (BVIL) Policies for Reliable LLM Tutoring Actions. *Published in Artificial Intelligence in Education (AIED 2026), Springer LNAI 16583.* [Paper]

IJCV 2026 Tajamul Ashraf, Rajes Manna, **Partha Sarathi Purkayastha**, Tavaheed Tariq, Janibul Bashir. Context Aware Grounded Teacher for Source Free Object Detection. *Published in the International Journal of Computer Vision (IJCV), 2026.* [Paper]

Accepted Workshop Papers

REALM @ EMNLP 2026 **Partha Sarathi Purkayastha**, Heejin Do. DySAS: Dynamic State-Aware Student Simulation Beyond Static Personas. *Accepted at the Research on Agent Language Models (REALM) Workshop, EMNLP 2026.* [Paper]

FAGEN @ ICML 2026 **Partha Sarathi Purkayastha**. When Thinking Before Retrieval Hurts: TraceBound Diagnostics for Adaptive Knowledge-Graph Retrieval. *Accepted at the Failure Modes in Agentic AI (FAGEN) Workshop, ICML 2026.* [Paper]

Awards

Jan'26 – Apr'26 **ASIA 2026 Data Challenge Track 2 Winner: Longitudinal ISNCSCI Imputation**.
○ Won American Spinal Injury Association challenge, Track 2, reconstructing full ISNCSCI neuroclassification sensory exams from longitudinal expedited E-ISNCSCI plus baseline ISNCSCI. Built leakage-safe pipeline for target-wise TabPFNv2.5 bagging with copy-through, symmetry, modality, and anchor post-processing.
Supervisor: Dr. Diego Paez-Granados

Experience

May'26 – Present **Research Assistant**, ETH SCAI Lab, Zürich / Nottwil, Switzerland.
○ Analyzing multimodal wearable and clinical data for **decongestion monitoring** in cardiac recovery.
Supervisor: Dr. Diego Paez-Granados

Sept'25 – Sept'26 **Research Intern**, ETH LRE Lab, Zürich, Switzerland.
○ Developed a **psychologically grounded, dynamic state-aware framework** for simulating **diverse and evolving student behaviours** beyond static personas, enabling rigorous evaluation of **adaptive LLM tutors** across multi-turn interactions. *Accepted at REALM @ EMNLP 2026.*
Supervisors: Prof.Mrinmaya Sachan, Dr.Heejin Do

Jul'24 – Jun'25 **Research Fellow**, Microsoft Research India, Bangalore, India.
○ Developed a **privacy taxonomy** and curated a **context-dependent summarization dataset**; evaluated **LLMs** using automatic metrics and human judgments, outperforming **GPT-4o** on privacy adherence while maintaining summary **consistency, relevance, and coherence**.
Supervisor: Dr.Akshay Nambi

Aug'23 – Jan'24 **Research Intern**, BITS Pilani, Pilani Campus, India.
○ Built a **fundus-image annotation tool** and collaborated with a hospital to curate a **3,000-image dataset**; applied image processing and custom deep-learning models to automate lesion annotation, achieving **over 80% accuracy and specificity**.
Supervisor: **Dr.Sundaresan Raman**

May'23 – Aug'23 **SWE Intern**, Uber, Hyderabad, India.
○ Developed an **end-to-end application** using **React.js**, **GraphQL**, and **GoLang** to manage **credit types** for **2000+ employees**. Migrated financial data to a **MySQL**-based centralized database for **secure transactions**. Automated integration and end-to-end testing, boosting system reliability and cutting onboarding time by **two months**

Jan'23 – May'23 **Research Intern**, Mercedes-Benz, Bangalore, India.
○ Enhanced component design by training **customized VAE models** for **parameter-controlled generation** on a **25,000+ point proprietary dataset**, and deploying an **application-style website** enabling user-friendly custom design generation for company research.
Supervisor: **Dr.Poonam Goyal**

Teaching Experience

Sept'25 – Jun'26 **AI and Digital Technology**, ETH Zürich, Teaching Assistant.
Supervisor: **Dr. Clara Meister**
Developing AI analytic initiatives for executive education in privacy-preserving ML and synthetic data projects

Jan'23 – Dec'23 **Techniques in Social Research**, BITS Pilani, Teaching Assistant.
Supervisor: **Dr.Tanu Shukla**
Introduced students to foundational research methodologies and assisted them in the analysis of social research projects

Jan'22 – May'22 **Microprocessors and Interfacing**, BITS Pilani, Teaching Assistant.
Supervisor: **Dr.Vinay Chamola**
Conducted lab sessions on MASM-based assembly programming, guiding students in hands-on interfacing exercises

Academic Service

2026 **Program Committee Member**: AI4GOOD @ ICML 2026, FAGEN @ ICML 2026 and REALM @ EMNLP 2026.

Community Activities

Oct'24-Nov'24 **Volunteer**, *Vision Empowerment*, Microsoft Research India, Bangalore, India.
Designed **Braille-based aids** to help **vision-impaired** children in basic **mathematical** learning through **haptic feedback**

Jan'22 – Dec'22 **Publicity Coordinator**, *Arunodoi*, BITS Pilani, India.
Led the **Media-editing** team for **Arunodoi**, promoting **Northeast India**'s cultural awareness and community engagement

Nov'21 – Jan'22 **Volunteer**, *Indian Red Cross Society*, BITS Pilani, India.
Volunteered for executing a **two-day long** Blood Donation Camp with **3000+** footfall and **800+** donations

Additional Projects

Sept'25 – Jan'26 **NeuroReach VR: Mixed-Reality Rehabilitation System for Stroke Recovery**.
○ Built a pass-through mixed-reality rehabilitation system on **Meta Quest 3S** with **Logitech MX Ink** for gross- and fine-motor training, clinician-readable event-level CSV logging, and a **10-user pilot study** achieving a **79.7/100** score on the **System Usability Scale (SUS)**.
Supervisors: **Marwen Mokni**, **Prof. Christoph Bauer**

Oct'23 – Dec'23 **Systemic Diabetic Retinopathy Analysis**.
○ Developed a **diabetic retinopathy** analysis portal focused on **systemic factors**. Achieved over **80% accuracy** in model testing through a comparative study using **TabNet** and **DeepForests** across hospitals in **India** and **Hong Kong**
Supervisor: **Dr.Sundaresan Raman**

Skills

Programing	Python, Java, TypeScript/JavaScript, C/C++, GoLang, SQL
Python Libraries	PyTorch, Transformers, Numpy, Pandas, Plotly, OpenCV
Development	Node.js, React.js/Next.js, GraphQL, Tailwind CSS
Tools	LaTeX, GitHub, Postman, Docker, MongoDB